Data as a Product: The New Paradigm

Treating data as a product (DaaP) requires the development of reliable systems that can be shared, reused and deliver value. Let's understand this concept and the important role open source is playing in its implementation.

Data is often nicknamed 'the new oil' as it requires processing similar to oil extraction before it reaches its useful state. It then becomes the power source for business decisions and product development. The technology behind artificial intelligence, automation, recommendation systems and digital ads relies fundamentally on data.

Data has evolved from being an operational byproduct to a product that companies now actively manage. The Data as a Product (DaaP) approach transforms business data management through open source tools such as Apache Kafka for real-time data streams and Airbyte for data integration.

What does 'Data as a Product' mean?

Data as a Product (DaaP) means treating data like a hardware or a software product. The data is actively managed, maintained and improved to serve end users, just like is done with any product. Data was previously locked away in centralised databases that were only accessible by IT teams, which limited users' ability to locate it. DaaP now puts the user, typically a data analyst, data scientist or developer, at the centre. This makes data more accessible, well-documented and reliable.

Today organisations use service level agreements (SLAs) for data management in the same way as they apply them to digital services. The agreements guarantee that data remains up to date, accurate and readily available. This transformation from static storage to delivering usable assets happens because data now serves multiple teams. Organisations across the board now view data as a product that requires reliability, maintenance and user-centred design.

Open source tools help businesses simplify their data management process. For example, Apache Airflow serves as a workflow automation tool which enables teams to construct and oversee data pipelines. It's like a scheduler for data tasks. The data build tool transforms unprocessed data into structured datasets ready for analysis. It functions as a data transformation system, which transforms raw data into meaningful reports and dashboard content. OpenMetadata enables teams in discovering data, performing quality checks and tracking data lineage. This builds trust and transparency because multiple users or teams access the same data.

From a product management point of view, DaaP involves thinking about who the users are, what problems they face and how data can solve those problems. However, from a data engineering perspective, it's about building systems and pipelines that deliver clean, consistent data at scale, often using open source stacks.

A wide range of sectors are embracing DaaP principles. Netflix relies on labelled behavioural data